

TOPIC IS SETS



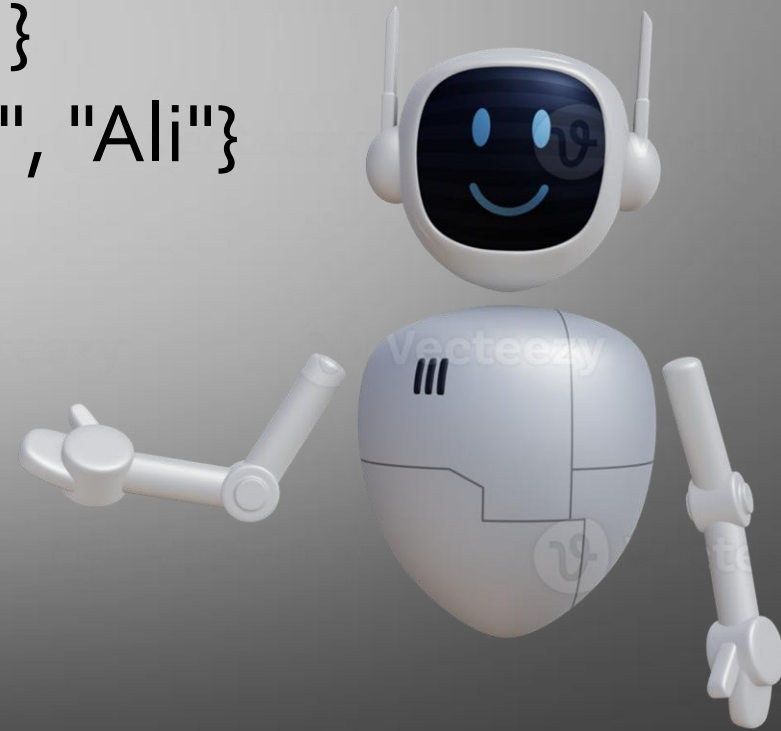
A set in Python is an unordered collection of unique elements, written using curly brackets `{ }`, where duplicate values are automatically removed. It is a built-in data type used to store multiple items in a single variable, and the elements inside a set do not have a fixed position or index. Because sets only keep unique values, they are especially useful when you want to remove duplicates from data or perform mathematical operations like union, intersection, and difference between collections.

EXAMPLE OF SETS

```
# Creating a set using curly brackets {}  
students = {"Ali", "Sara", "John", "Sara", "Ali"}  
print("Set of students:", students)
```

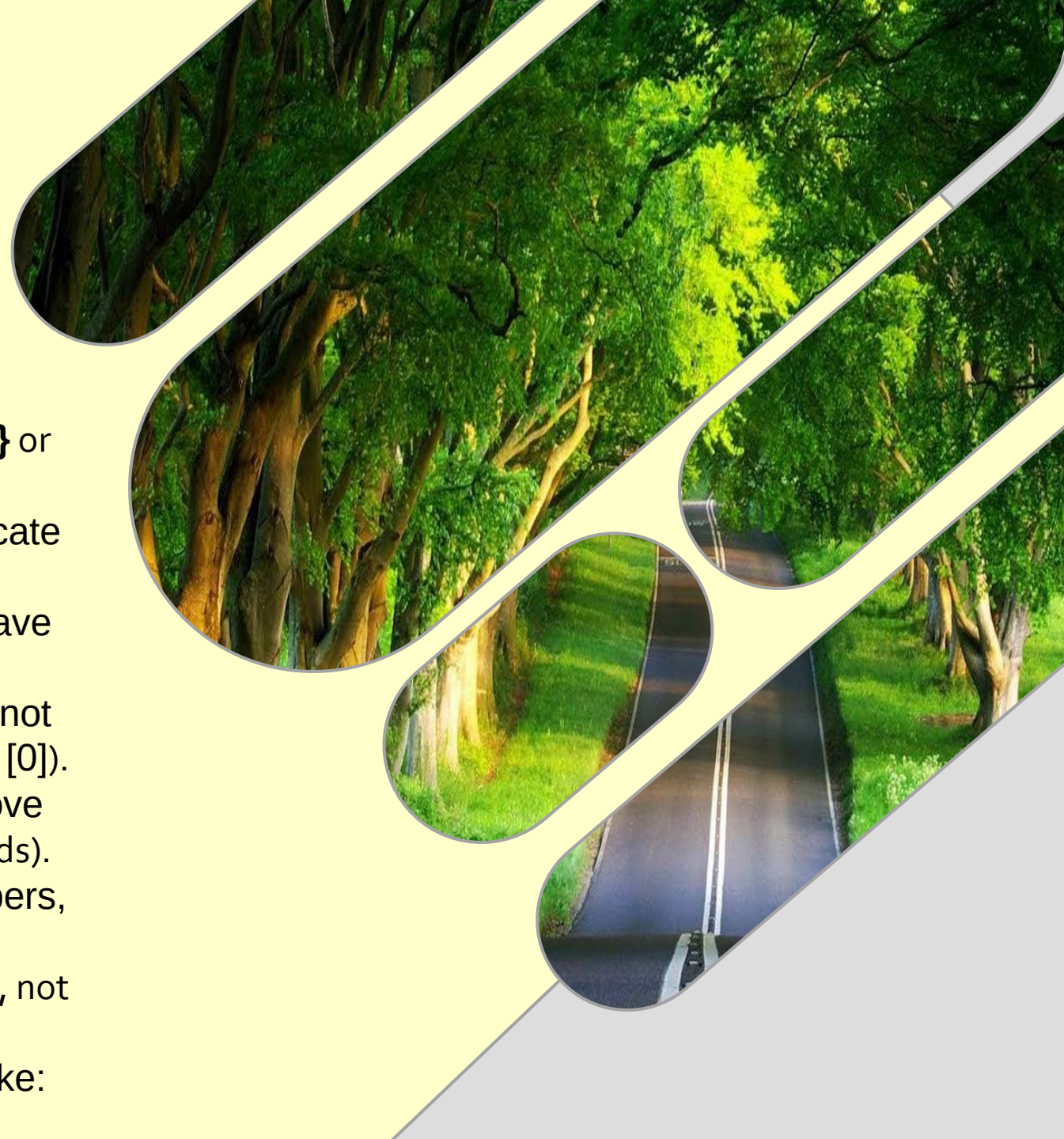
output :-

Set of students: {'Ali', 'Sara', 'John'}



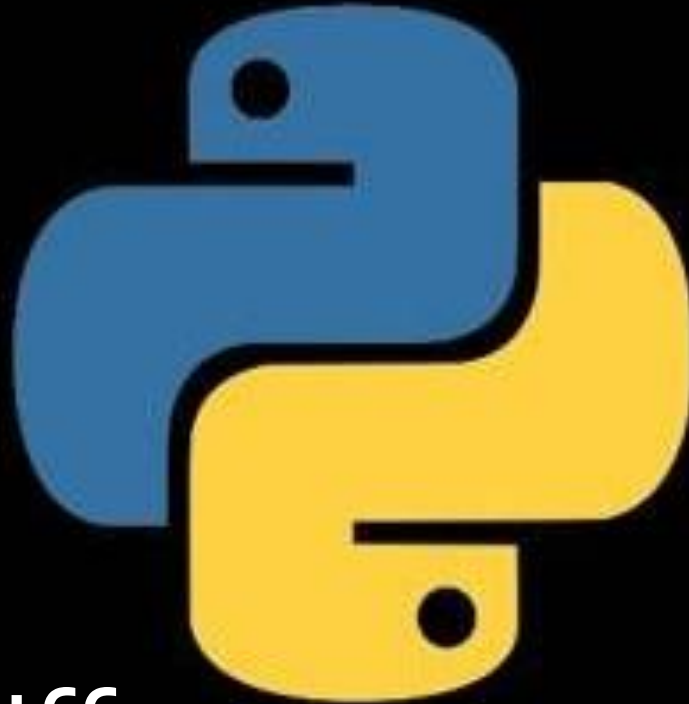
➤ SOME IMPORTANT POINTS FOR SETS

- Sets are written using **curly brackets { }** or the `set()` function.
- Sets store **only unique elements** (duplicate values are automatically removed).
- Sets are **unordered** (elements do not have a fixed position).
- Sets do **not support indexing** (you cannot access elements using index numbers like `[0]`).
 - Sets are **mutable** (you can add or remove elements using `add()` and `remove()` methods).
- Sets can store different data types (numbers, strings, etc.).
- An empty set must be created using `set()`, not `{ }` (because `{ }` creates a dictionary).
- Sets support mathematical operations like:



SETS OF OPERATION

1. Union
2. Intersection
3. Difference
4. Symmetric Difference

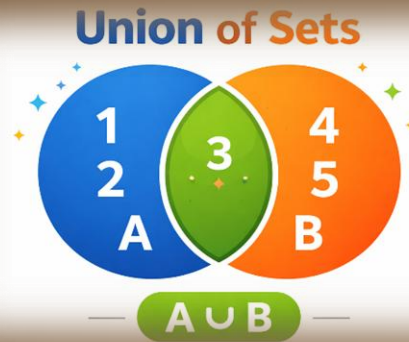


➤ UNION IN PYTHON SETS

■ Definition :

- **Union** is a set operation that combines all the unique elements from two or more sets into a single set. If there are common (same) elements in the sets, they appear only once in the result because a set does not allow duplicate values.

Example :- We have a best example for this union set



A = {1, 2, 3}

B = {3, 4, 5}

```
result = A | B  
print(result)
```

Output :

{1, 2, 3, 4, 5}

➤ INTERSECTION IN PYTHON SETS

- Definition :
- **Intersection** is a set operation that finds the common (same) elements of two or more sets. That is, only the elements that are present in all the sets appear in the result.

Example :- We have a best example for this intersection sets

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

```
Common_elements = A & B  
print(Common_elements)
```

Output :
 $\{3, 4\}$

➤ DIFFERENCE IN PYTHON SETS

■ Definition :

- **Difference** is a set operation that returns the elements of the first set that are not present in the second set.
That is, the result contains only the unique elements of the first set that do not exist in the second set.

Example :- We have a best example for this Difference sets

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

```
diff = A - B  
print(diff)
```

Output :
 $\{1, 2\}$



Unique Elements of First Set
DIFFERENCE

➤ SYMMETRIC DIFFERENCE IN PYTHON SETS

- Definition :
- **Symmetric Difference** is a set operation that returns the elements of two sets that are present in only one set, removing the elements that are common to both sets.

Example : we have a best for this symmetric difference set

A = {1, 2, 3, 4}

B = {3, 4, 5, 6}

```
sym_diff = A ^ B  
print(sym_diff)
```

Output :
{1, 2, 5, 6}





THANK
YOU